WATER RESOURCES RESEARCH, VOL. 25, NO 2, PAGES 152–160, FEBRUARY 1989

Is Optimization Optimistically Biased?

BENJAMIN F. HOBBS

Departments of Systems Engineering and Civil Engineering, Case Western Reserve University, Cleveland, Ohio

ANN HEPENSTAL

Procter and Gamble, Cincinnati, Ohio

Does optimization systematically lead to solutions that appear better than they actually turn out to be when implemented? The answer can be yes if there are errors in estimating objective function coefficients. Even if such errors are unbiased, the calculated value of the objective function for the optimal solution will, in an expected value sense, overstate that solution's true performance. This presupposes that errors in the constraint set are relatively unimportant. The existence of such a bias is shown by proof; Monte Carlo simulations of two realistic water resources optimization problems show its significance for water planners. The most important implication is that the estimated net benefits of model solutions may be exaggerated compared to existing water systems, whose performance is generally known with more accuracy.

INTRODUCTION

Optimization is the search for a "best" solution using mathematical methods. Optimization problems can be phrased generally as

$$\max_{\mathbf{X} \epsilon F} z(\mathbf{X} \mid \mathbf{C}) \tag{1}$$

where $\mathbf{X}$ is a vector of decision variables, $z(\mathbf{X} \mid \mathbf{C})$ an objective function with coefficients $\mathbf{C}$, and F is the feasible region. $\mathbf{X}^*(\mathbf{C})$ is the optimal solution.

Methods for solving such problems are in the analytical tool kit of many water resources engineers. This and other water journals have published numerous papers on optimization presenting both theoretical advances and applications (for surveys, see *Friedman et al.* [1984], *Loucks et al.* [1984], *Rogers* [1980], and *Rogers and Fiering* [1986]). Although there is debate on the true utility of optimization [*Liebman*, 1976; *Rogers and Fiering*, 1986; *Wyatt et al.*, 1988], every water curriculum worthy of the name now includes at least one course on the topic.

Optimization has several uses in water resources. One is to help find better designs and operating policies and to estimate the benefits of their implementation. The above surveys cite many examples, including the location and sizing of reservoirs, operation of hydropower systems, design of remedial actions for groundwater contamination, and siting and sizing of wastewater plants, distribution lines, and sanitary sewers. A second use is in statistical estimation. A third is policy analysis: the study of the impacts of changes in government policy upon a system, such as a market, which is presumed to behave in an optimal fashion [e.g., *Hobbs and Meier*, 1979]. However, solutions to such models are never implemented unaltered [*Loucks et al.*, 1984, pp. 221–222]:

> . . . most would agree that applications of modeling designed to help or aid decision-makers should only rarely, if at all, be designed to provide actual decisions regarding often complex management or policy problems the model output serves only to focus the debate about what to do; it is not itself a substitute for the planning, managing, or policy making process.

Model solutions are just one of many inputs to the decision process. Solutions are modified to reflect institutional considerations and are often subjected to extensive simulation. Or optimization may be used just to screen out obviously inferior solutions. Optimization can also yield insights into the workings of the system, the critical issues, and the general form of good solutions.

In this paper, we focus on the use of optimization to predict the performance of alternative designs, operating rules, and government policies. We address the following question, Do estimation errors in objective function coefficients cause optimization to have a fundamental bias toward optimism? Specifically, will such errors lead us to expect to gain more than we really will when we implement the optimal solution? Below, we show that under certain circumstances, the answer is yes.

In mathematical terms, let $\hat{\mathbf{C}}$ be an estimate of the true coefficients $\mathbf{C}_T$:

$$\hat{\mathbf{C}} = \mathbf{C}_T + \mathbf{e} \tag{2}$$

where $\mathbf{e}$ is a vector of unbiased estimation errors $E(\mathbf{e})=0$. Let $z(\mathbf{X}^*(\mathbf{C}_a) \mid \mathbf{C}_b)$ be the value of the objective, calculated using $\mathbf{C}_b$, of the optimal solution obtained using $\mathbf{C}_a$. Our question is whether, for a maximization problem

$$E(z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \hat{\mathbf{C}})) > E(z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \mathbf{C}_T)) \tag{3}$$

"optimistic bias"

That is, is the expected value of the calculated objective function better than the expected true performance of the selected alternative?

If optimization models are indeed overly optimistic, we should be wary when using optimization to predict how much improvement we can obtain in a system. Modelers often enthusiastically point out how much better their solutions would be compared to some status quo [e.g., *Esmaeil-Beik and Yu*, 1984; *Leighton and Shoemaker*, 1984]. But an optimistic bias might cause a model's solution to appear better than present designs or policies even if, in truth, it is not.

The plan of the paper is as follows. After a literature survey, we prove, under reasonable conditions, that unbi-

Paper number 88WR03708
0043-1397/89/88WR-03708$05.00

ased estimation errors in objective function coefficients lead to an optimistic bias (in an expected value sense) for a very general class of optimization models. An important assumption underlying this is that errors in constraints are relatively insignificant. The last part of the paper demonstrates the optimistic bias for two realistic water resources planning problems: a multiobjective basin planning problem and a nonlinear program for assessing water supply and use alternatives for power plants. An appendix examines the implications of errors in constraint coefficients.

LITERATURE SURVEY

This work builds upon three bodies of literature on the effect of parameter error in mathematical programs. Decision theorists have been concerned with the explicit incorporation of uncertainties in decision making [e.g., *Raiffa and Schlaifer*, 1961; *Krzysztofowicz*, 1983]. Several water researchers have examined the effect of parameter errors on solutions of deterministic models. Finally, a classic problem in operations research has been to determine the probability distribution of optimal objective functions under random parameters.

To decision theorists, the correct phrasing of the objective function (1) is a stochastic one (R. Krzysztofowicz, personal communication, 1987):

$$\max_{\mathbf{X}} E(z(\mathbf{X} \mid \mathbf{C}_T) \mid \hat{\mathbf{C}}) = \int z(\mathbf{X} \mid \mathbf{C}_T) f(\mathbf{C}_T \mid \hat{\mathbf{C}}) \, d\mathbf{C}_T \qquad (4)$$

That is, the objective is to maximize the expected true value of z. This assumes that $\mathbf{C}_T$ is the only unknown set of parameters and that decision makers are risk neutral. $\hat{\mathbf{C}}$ is treated as an experimental observation and $\mathbf{C}_T$ as uncertain. The $f(\mathbf{C}_T \mid \hat{\mathbf{C}})$ is a multivariate posterior probability density, which should be calculated from a multivariate prior $f(\mathbf{C}_T)$ and a multivariate conditional probability distribution $f(\hat{\mathbf{C}} \mid \mathbf{C}_T)$ using Bayes Law. The optimal solution to this
(!) problem $\mathbf{X}^+(\hat{\mathbf{C}})$ is not necessarily the same as $\mathbf{X}^*(\hat{\mathbf{C}})$, the solution to the deterministic problem (1). Further, by construction, the value of the objective function calculated using (4) is an unbiased estimator of the true value $z(\mathbf{X}^+(\hat{\mathbf{C}}) \mid \mathbf{C}_T)$. However, as we show below, the objective function value yielded by (1) is, in general, biased. Thus from the decision-theoretic point of view, the optimistic bias of (1) results of an improper model of the decision problem—the use of deterministic representations of a reality that is intrinsically uncertain (R. Krzysztofowicz, personal communication, 1987).

Water researchers have analysed the costs that result from errors in deterministic water system optimization models. *Thomas* [1967] analyzed two types of costs caused by parameter error. One type, which he calls "type A error" or

type A error

"loss of economic efficiency," equals $[z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \mathbf{C}_T) - z(\mathbf{X}^*(\mathbf{C}_T) \mid \mathbf{C}_T)]/z(\mathbf{X}^*(\mathbf{C}_T) \mid \mathbf{C}_T)$. This is the realized objective function value minus the value which could have been obtained had $\mathbf{C}_T$ been known and the true optimum chosen, expressed as a fraction of the latter. The negative of this error is known as "regret" [e.g., *Bell*, 1982]. Note that if there is any chance that an alternative which is not the true optimum will be chosen, then the expected regret must exceed 0 for maximization problems (assuming that $z > 0$).

connection to regret

Thomas also defines a "type B error," equaling $[z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \hat{\mathbf{C}}) - z(\mathbf{X}^*(\mathbf{C}_T) \mid \mathbf{C}_T)]/z(\mathbf{X}^*(\mathbf{C}_T) \mid \mathbf{C}_T)$. Others call this "ex ante perceived resource loss" [*Rogers*, 1978] and "misrepresentation of minimal costs" [*Loucks et al.*, 1981]. In words, it is the difference between the estimated objective function and the true worth of the true optimum. Below (equation (13)), we prove, under mild restrictions, that estimation errors in objective function coefficients cause the expectation of the type B error to be strictly positive (assuming that z is to be maximized and $z > 0$). Equation (3), (!!) which might be called "disappointment" [*Bell*, 1985], is the difference between Thomas's type B and A errors, multiplied by $z(\mathbf{X}^*(\mathbf{C}_T) \mid \mathbf{C}_T)$.

type B error

Thomas [1967] then calculates the type A and B errors for a simple capacity expansion problem for a variety of cases. He found that the type A error, which represents the real resource loss, is the much smaller of the two. He also examines the relative sizes of errors arising from technological versus socioeconomic uncertainties. *Rogers* [1978] builds on this analysis by examining costs caused by errors resulting from incorrect model form.

Elsewhere in the water literature, *Read and Boshier* [1986] conclude that some deterministic reservoir operation models overestimate benefits. The reason is that those models are overoptimistic about the ability of management to cope with random inflow variations.

In the operations research literature, a classic problem of stochastic programming is the "distribution problem." The challenge there is to find the probability distribution of $z(\mathbf{X}^*(\mathbf{C}) \mid \mathbf{C})$, given that $\mathbf{C}$ is random [*Wets*, 1980].

Several workers have established bounds for the mean of $z(\mathbf{X}^*(\mathbf{C}) \mid \mathbf{C})$. *Mangasarian* [1964] found that $z(\mathbf{X}^*(\mathbf{C}) \mid \mathbf{C})$ is a convex function of $\mathbf{C}$ for any form of feasible region $\mathbf{F}$ if $z(\;)$ is convex in $\mathbf{C}$ for any fixed feasible value of $\mathbf{X}$. This implies that $E(z(\mathbf{X}^*(\mathbf{C}) \mid \mathbf{C})) \geq z(\mathbf{X}^*(E((\mathbf{C})) \mid E(\mathbf{C}))$, if $z(\;)$ is to be maximized. He has also established the following bounds:

$$E(z(\mathbf{X}^*(\mathbf{C}) \mid \mathbf{C})) \geq \max_{\mathbf{X}} E(z(\mathbf{X} \mid \mathbf{C}))$$

$$\geq E(z(\mathbf{X}^*(E(\mathbf{C})) \mid \mathbf{C})) \geq z(\mathbf{X}^*(E(\mathbf{C})) \mid E(\mathbf{C})) \qquad (5)$$

If we substitute $\hat{\mathbf{C}}$ for $\mathbf{C}$ and $\mathbf{C}_T$ for $E(\mathbf{C})$ in this inequality, and note that by definition of optimality, $z(\mathbf{X}^*(E(\mathbf{C})) \mid E(\mathbf{C})) \geq$? def of optimality ?
$z(\mathbf{X}^*(\mathbf{C}) \mid E(\mathbf{C}))$, we get (3), except that the inequality is not a strict one. The paper by *Bard and Chatterjee* [1985] implies that for LPs, the inequalities of (5) hold as equalities only if the optimal solution under $E(\mathbf{C})$ is optimal under all $\mathbf{C}$ which have nonzero probability. In the next section, we establish the same condition for a more general class of mathematical programs. Later work has put bounds upon both $z(\mathbf{X}^*(\mathbf{C}) \mid \mathbf{C})$ and $E(z(\mathbf{X}^*(\mathbf{C}) \mid \mathbf{C}))$, given known bounds upon possible realizations of $\mathbf{C}$ [e.g., *Bard and Chatterji*, 1985].

Itami [1974] shows that the expected objective function value of a linear program with random $\mathbf{C}$ is nondecreasing as the variance of $\mathbf{C}$ increases and that, due to duality, the opposite is true of the right-hand sides of the constraint set. We interpret his result as implying that the degree of optimistic bias (3) will be nondecreasing in the variance of the estimation errors.

In our search of the literature, we found no application of these results to estimation errors; in particular, the implications for the optimistic biasedness of optimization have been missed. An example of the type of conclusion that has instead been drawn is that of *Mangasarian* [1964]: "wait and see" decisions, in which $\mathbf{X}$ is chosen after $\mathbf{C}$ is realized, are

simple explanation for "wait and see" bound/decisions better than "here and now" decisions, in which $\mathbf{X}$ must be picked before $\mathbf{C}$ is known.

DEMONSTRATION OF THE RESULT

In this section, we prove, under certain reasonable conditions, that if z is to be maximized and is linear in the coefficients $\mathbf{C}$, then $E(z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \hat{\mathbf{C}})) > E(z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \mathbf{C}_T))$ (equation (31). If z is also linear in $\mathbf{C}$, then (3) becomes

conditions

$$E(\hat{\mathbf{C}}'\mathbf{X}^*(\mathbf{C})) > E(\mathbf{C}_T{}'\mathbf{X}^*(\hat{\mathbf{C}})) \tag{6}$$

D. Maidment and M. Fiering (personal communications, 1987) point out that the left side of (6) does not, in general, equal $E(\hat{\mathbf{C}})'E(\mathbf{X}^*(\hat{\mathbf{C}}))$ (and thus if $E(\mathbf{e})=0$, $\mathbf{C}_T{}'E(\mathbf{X}^*(\hat{\mathbf{C}}))$) because $\hat{\mathbf{C}}$ and $\mathbf{X}^*(\hat{\mathbf{C}})$ are dependent. Let $\Delta\mathbf{X}$ equal $\mathbf{X}^*(\hat{\mathbf{C}})$ minus $\mathbf{X}^*(\mathbf{C}_T)$. Analyzing the left side of (6) and assuming that $E(\mathbf{e}) = 0$ yields

$$E(\hat{\mathbf{C}}'\mathbf{X}^*(\hat{\mathbf{C}})) \triangleq E([\mathbf{C}_T + \mathbf{e}]'[\mathbf{X}^*(\mathbf{C}_T) + \Delta\mathbf{X}^*])$$

equal by definition

$$= E(\mathbf{C}_T{}'[\mathbf{X}^*(\mathbf{C}_T) + \Delta\mathbf{X}^*]) + E(\mathbf{e}'\Delta\mathbf{X}^*) \tag{7}$$

This is the same as (6), except for the last term on the right side. This states that there is an optimistic bias only if
(!!) $E(\mathbf{e}'\Delta\mathbf{X}^*)$ exceeds zero; that is, only if the errors are positively correlated with the changes they cause. An example of this correlation in a math program is given later in this paper.

The assumptions of the proof are as follows.

Assumption 1. The optimization problem is of the form

$$\max_{\mathbf{X}\,\varepsilon F} z(\mathbf{X} \mid \mathbf{C}) = \Sigma_i C_i z_i(\mathbf{X}) \tag{8}$$

where F is a bounded feasible region with two or more distinct values of $\mathbf{X}$, and $z_i(\mathbf{X})$ is any function of the decision variables $\mathbf{X}$ which is bounded for $\mathbf{X}\varepsilon F$; $z_i(\ \ )$ may be convex or concave.

We assume that only $\mathbf{C}$ is subject to estimation error; as the appendix shows, estimation errors in F can in some cases imply optimism and in other case imply pessimism. This general math program encompasses nearly all types of optimization models used in water planning; as examples, the special cases of linear and quadratic programming (if the $z_i(\ \ )$ are linear or quadratic functions of $\mathbf{X}$, respectively) and integer programming (if F is defined so that some variables are only integer valued). Without loss of generality, we assume that the solution $\mathbf{X}^*(\mathbf{C}_T)$ to (8) under the true parameters $\mathbf{C}_T$ is unique.

Assumption 2. The decision process consists of getting the solution (8) and then implementing it, at which time its true performance is revealed. The calculated value of the objective function $z(\mathbf{X}^*(\hat{\mathbf{C}}))$ is used to predict the performance of the optimal solution $\mathbf{X}^*(\hat{\mathbf{C}})$. This is perhaps a caricature of the actual role of optimization in decision making, in that solutions are never implemented without subsequent simulation and modification. But like any good caricature, this one captures certain essential characteristics of the subject—here, the choosing of some alternatives over others based on a performance criterion. Any bias resulting from optimization is likely to persist through subsequent modifications of solutions unless independent information sources are used to check the original net benefit estimates.

Assumption 3. Let there be estimation errors in $\mathbf{C}$ such that $E(\mathbf{e}) = 0$ and

$$\hat{\mathbf{C}} = \mathbf{C}_T + \mathbf{e} \tag{9}$$

Assumption 4. Let the possible realizations of $\hat{\mathbf{C}}$ be divided into two sets:

$$A = \{\hat{\mathbf{C}} \text{ such that } \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})] = \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)]\} \tag{10a}$$

$$B = \{\hat{\mathbf{C}} \text{ such that } \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})] > \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)]\} \tag{10b}$$

That is, set A consists of events such that $\mathbf{X}^*(\mathbf{C}_T)$ is still optimal under $\hat{\mathbf{C}}$, and B represents values of $\hat{\mathbf{C}}$ in which $\mathbf{X}^*(\mathbf{C}_T)$ is suboptimal. We assume that $\int_{\hat{\mathbf{C}}\varepsilon B} f(\hat{\mathbf{C}})\, d\hat{\mathbf{C}} > 0$, where $f(\hat{\mathbf{C}})$ is the probability density of $\hat{\mathbf{C}}$; that is, there is a positive probability of obtaining a $\hat{\mathbf{C}}$ in set B. This assumption is not unreasonable, particularly for realistically sized mathematical programs. This is borne out by our experience in conducting parametric programming analyses of moderately sized (50 or more variables and constraints) linear programs. Generally, changes in $\mathbf{C}$ that are well within the margin of estimation error lead to changes in the optimal basic solution.

The proof proceeds as follows. Since $\mathbf{X}^*(\mathbf{C}_T)$ is by definition feasible,

$$\Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})] \geqslant \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)] \tag{11}$$

Then, by taking expectations and invoking the assumption $E(\mathbf{e})=0$, we have

$$E(\Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})]) \geqslant E(\Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)])$$

$$= \Sigma_i C_{Ti} z_i[\mathbf{X}^*(\mathbf{C}_T)] \tag{12}$$

which is a rephrasing of *Mangasarian*'s [1964] result.

If, as we assume, set B has a nonzero probability, then (12) holds as a strict inequality (as does equation (3)). This can be proven as follows:

$$E(z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \hat{\mathbf{C}})) = E(\Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})])$$

$$= \int_{\hat{\mathbf{C}}} \{f(\hat{\mathbf{C}})\ \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})]\}\, d\hat{\mathbf{C}}$$

$$= \int_{\hat{\mathbf{C}}\varepsilon A} \{f(\hat{\mathbf{C}})\ \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})]\}\, d\hat{\mathbf{C}}$$

$$+ \int_{\hat{\mathbf{C}}\varepsilon B} \{f(\hat{\mathbf{C}})\ \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})]\}\, d\hat{\mathbf{C}}$$

and, by definition of sets A and B:

$$= \int_{\hat{\mathbf{C}}\varepsilon A} \{f(\hat{\mathbf{C}})\ \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)]\}\, d\hat{\mathbf{C}}$$

$$+ \int_{\hat{\mathbf{C}}\varepsilon B} \{f(\hat{\mathbf{C}})\ \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\hat{\mathbf{C}})]\}\, d\hat{\mathbf{C}}$$

$$> \int_{\hat{\mathbf{C}}\varepsilon A} \{f(\hat{\mathbf{C}})\ \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)]\}\, d\hat{\mathbf{C}}$$

TABLE 1. Santa Cruz Basin Attributes*

Attribute	Units	Worst*	Best*	Rescaling	Weight
x_1, aquifer level change	ft/year	3.5	1.6	0–150	9
x_2, urban water quality	subjective	e	a	0–110	3
x_3, agricultural water quality	subjective	e	a	0–130	3
x_4, expected flood losses	dollars	26.33	0	0–160	4
x_5, expected flood frequency	number/year	0.04	0.003	0–160	5
x_6, designated environmental area preservation	subjective	e	a	0–150	5
x_7, effect on wildlife and vegetation	subjective	e	a	0–200	5
x_8, implementation costs	present dollars	36.6	0	0–200	2
x_9, operation and management costs	present dollars	38.2	0	0–110	2
x_{10}, indirect costs	subjective	e	a	0–100	2
x_{11}, natural resources use	subjective	e	a	0–100	2
x_{12}, preservation of existing recreational facilities	subjective	e	a	0–100	1.5
x_{13}, creation of new recreational opportunities	subjective	e	a	0–76	1.5

Source: *Gershon et al.* [1982]. One foot = 30.48 cm.
*For subjective scales, five categories (a, b, c, d, and e) are defined.

$$+ \int_{\hat{\mathbf{C}} \in A} \{f(\hat{\mathbf{C}})\ \Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)]\}\ d\hat{\mathbf{C}}$$

$$= E(\Sigma_i \hat{C}_i z_i[\mathbf{X}^*(\mathbf{C}_T)]) = \Sigma_i C_{Ti} z_i[\mathbf{X}^*(\mathbf{C}_T)]$$

$$> E(\Sigma_i C_{Ti} z_i[\mathbf{X}^*(\hat{\mathbf{C}})]) = E(z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \mathbf{C}_T)) \tag{13}$$

The last inequality results from the assumption that the last inequality results from the assumption that $\mathbf{X}^*(\mathbf{C_T})$ is unique and the assumption that set B has nonzero probability. This completes the proof of (3) for z linear in $\mathbf{C}$.

proof of (3) also holds for nonlin sys if ...

R. Krzystofowicz (personal communication, 1987) has pointed out that this proof in general applies to any $z(\mathbf{X} \mid \mathbf{C})$ for which the $E(z(\mathbf{X} \mid \hat{\mathbf{C}})) = z(\mathbf{X} \mid \mathbf{C}_T)$.

Illustrations of the Optimistic Bias

To show some of the implications of the bias, we present three examples: a simple assignment problem, a multiobjective river basin planning problem, and a nonlinear water supply and use programming model. Estimates of the bias for the last two examples are obtained by Monte Carlo simulation. Details on the simulations are available from the authors.

A Simple Example

The optimistic bias is shown here for an assignment linear program:

$$\text{Max} \sum_j C_j x_j \tag{14}$$

Subject to

$$\sum_j x_j = 1 \tag{15}$$

$$x_j \geq 0, \ \forall\ j \tag{16}$$

In words, find which alternative j should be implemented ($x_j = 1$) in order to maximize benefits. As in the previous section, the objective function coefficient estimates $\hat{C}_j$ are subject to random estimation errors e_j, $E(e_j)=0$;

$$\hat{C}_j = C_{Tj} + e_j \tag{17}$$

Further, assume that the e_j are independently identically distributed (iid) random variables. This assumption is not necessary, for as long as the errors are not perfectly (and positively) correlated, (3) will still hold, in general. Finally, as an extreme case, assume that $C_{Tj} = 1, \forall\ j$. Under these assumptions, the problem of finding $E[z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \hat{\mathbf{C}}) - z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \mathbf{C}_T)] = E(z(\mathbf{X}^*(\hat{\mathbf{C}}) \mid \hat{\mathbf{C}})) - 1$ is a problem in order statistics: find the expected value of the largest of n iid random variables and then subtract $C_{Tj} = 1$, where n is the number of variables x_j. If e_j is uniformly distributed over the range [−0.5, +0.5], then the expected value of the highest $\hat{C}_j$ is $0.5 + n/(n + 1)$. This results in an expected optimistic bias of $n/(n + 1) - 0.5$, which is positive for $n > 1$.

Even if the C_{Tj} are not equal, (3) still holds. The only exception occurs if the probability distribution of $\hat{C}_j$ for the j with the highest C_{Tj} does not overlap probability distributions for any of the other $\hat{C}_j$; that is, if set B (equation (10*b*)) has zero probability. (!!)

Discrete Multiobjective Optimization Example

Here, we simulate a discrete multiobjective river basin planning problem. *Gershon et al.* [1982] evaluated river basin development plans for the Santa Cruz Basin of Arizona. In their study, 25 options for water supply, flood control, and other purposes were compared on the basis of 13 attributes, defined in Table 1. (We also conducted a simulation of the *Keeney and Wood* [1977] multiobjective river basin study, which confirmed the conclusions below.) This type of multiobjective analysis is used in two stages of decision making: first, to screen a large number of alternative plans, leaving a subset for later, more detailed study; and second, to help select a final choice from among a handful of alternatives that have been analyzed extensively.

Gershon et al. [1982] applied several multiobjective methods; for simplicity, we adopt an additive utility function. In

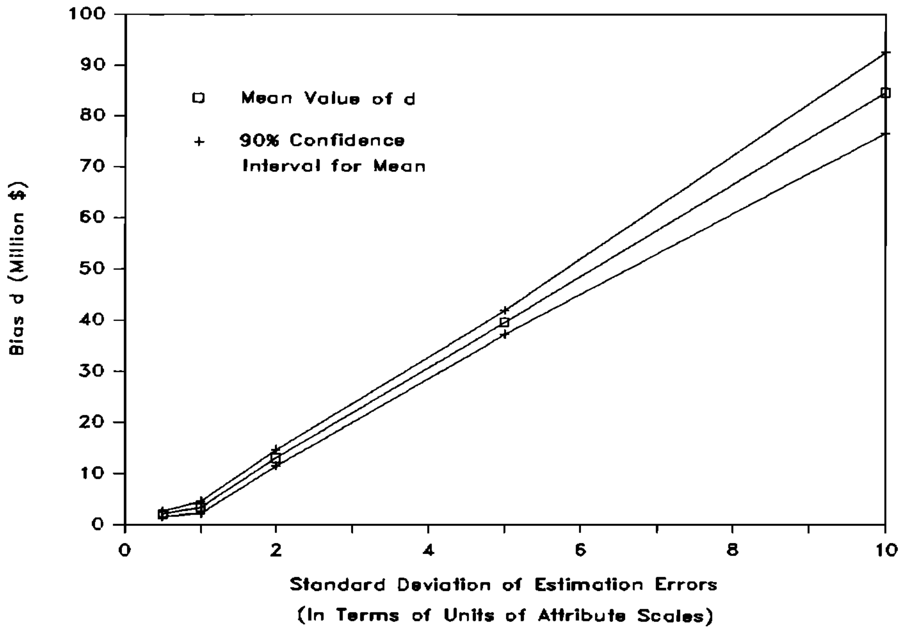

Fig. 1. Decrease in true cost needed to equate true and estimated utilities, Santa Cruz Basin.

this method, the option that maximizes the weighted sum of single-attribute utility functions is picked as "best":

$$\operatorname*{Max}_k U(\mathbf{X}_k) = \operatorname*{Max}_k \sum_j w_j u_j(x_{jk}) \tag{18}$$

where

$U(\mathbf{X}_k)$ overall utility of alternative k;
x_{jk} value of attribute j for alternative k;
$u_j(\ )$ single attribute utility function converting attribute j into a measure of utility;
w_j weight of attribute j (Table 1).

We also conducted simulations using multiplicative utility functions [*Keeney and Wood*, 1977] and compromise programming; the conclusions are the same.

For each basin, we randomly generate a series of problems and then apply the additive rule to find the best option in each problem. We then compare the estimated utility of the best option to its assumed "true" utility. In each problem, the values of x_{jk} we use (denoted as $\hat{x}_{jk}$) are generated as follows:

$$\hat{x}_{jk} = x_{Tjk} + e_{jk} \tag{19}$$

The x_{Tjk} we use as the true values are those assumed by *Gershon et al.* [1982]. The e_{jk} are generated as normal variables, with following correlations: $r(e_{jk}\, e_{jm}) = 1/3$ for $k \neq m$ and $r(e_{jk}\, e_{nk}) = 1/3$ for $j \neq n$. The standard deviations, in terms of points on the attribute scales (Table 1), are the same for all attributes. This error structure reflects the likelihood that estimates of the attributes of a given option would have similar biases, as might the estimates of one attribute across different options.

The measure of method "optimism" we select is the decrease d in true economic cost (x_{T8}) in the chosen alternative h necessary to make its true utility equal to its estimated utility:

$$w_8 u_8(x_{T8h} - d) + \sum_{j \neq 8} w_j u_j(x_{Tjh}) = \sum_j w_j u_j(\hat{x}_{jh}) \tag{20}$$

This quantifies in monetary terms how much worse the alternative truly is compared to its estimated (error-laden) utility. If there is an optimistic bias, d will be positive, on average; that is, the true cost would have to improve (decrease) in order to equate the true and estimated utilities.

We undertake simulations in order to find out how large the errors would have to be to induce a significant bias. Figure 1 shows the results, displaying, as a function of error standard deviation, the mean cost decrease d needed in the true evaluation in order to equate the true and estimated utilities. Also given are 90% confidence limits for the mean d (each based on a sample of at least 60). As expected, the mean d is significantly positive, even for error standard deviations as tiny as 0.5% of the range of the attributes.

The importance of this optimistic bias can be assessed by contrasting the mean d with the overall range of the cost attribute. The mean bias is as large as \$85 $\times$ 10^6, which is over twice the range of implementation cost. Thus if costs are important in these problems, so too is the potential optimistic bias.

What, in terms of a real resource loss, does this bias mean? Perhaps little, if such a multiobjective analysis is used just to conduct a preliminary screening of a large number of alternatives. In the subsequent, final decision stage, the surviving alternatives would be scrutinized in much more detail, with additional information coming from field studies and other primary sources. If these sources of information are independent of the data used during the screening phase, then those alternatives whose benefits are most biased are likely to be "found out" and the biases erased. However, if the same sources of information are used in both the screening and final analysis stages, then an optimistic bias that occurs during screening might persist. As an example, consider a water supply project in a developing country. Errors in projections of foreign exchange costs may bias a screening exercise towards a particular type of alternative and exaggerate its net benefits. If, in the final decision stage, estimates of those costs are obtained from the same source

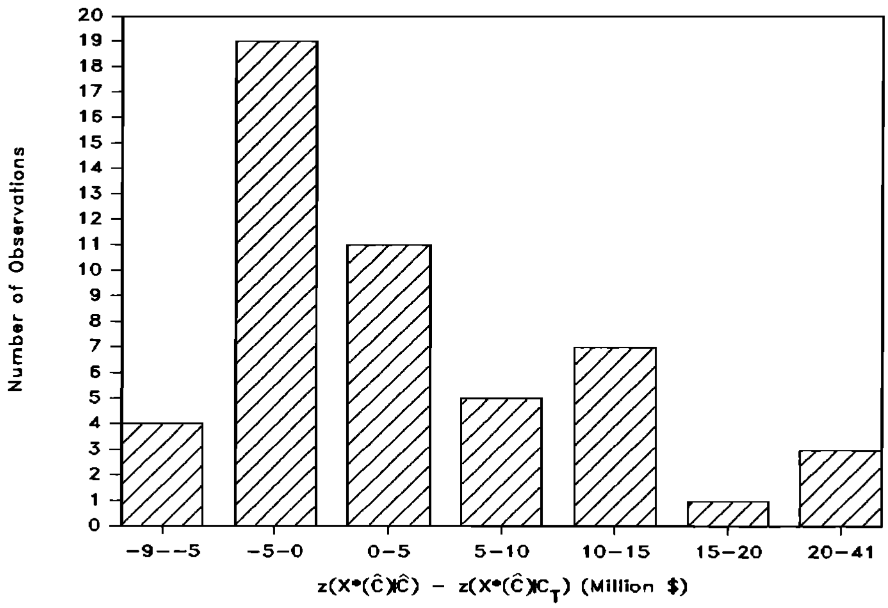

Fig. 2. Histogram, estimated objective function minus true objective function function, Brazos River NLP.

as during the screening (e.g., the World Bank), then this bias would not be corrected.

An optimistic bias could cause a real resource loss if present in the final decision stage. A project whose net benefits, as measured by the attributes, are exaggerated due to this bias is more likely to be able to compete successfully for the limited monies available in a funding agency's budget. This bias may convince an agency to sponsor a project which, after implementation, is found to provide fewer net benefits than the system it replaced. The result is a real loss to society: the loss of the benefits of the replaced system and the opportunity cost resulting from diversion of funds from other possible projects.

Mathematical Programing Example

The model we analyze is a nonlinear program (NLP) extracted from a model of power plant water supply and use in the Texas-Gulf region [*Hobbs*, 1984]. The NLP is a simplified representation of water supplies and conservation options available to electric utilities in the Brazos River basin of Texas during the year 2000. It imitates cost-minimizing choices by utilizes among water sources (unallocated surface and groundwater, water rights transfers) and cooling technologies (evaporative cooling, mixed wet/dry cooling, dry cooling). Its purpose is to guage the potential benefits of innovative mixed wet/dry and dry cooling technologies, although it could also be used for water supply planning.

The objective function represents the annual benefits of water consumption, in terms of the avoided cost of conservation measures, minus the expense of aqueducts, groundwater supply, water rights purchases, and reservoirs. The objective is linear, except for the cost of water rights, which is quadratic. Constraints define locations of power plants, water use under different cooling technologies, and availabilities of various sources of water. The NLP has 30 variables representing uses and supplies of water.

The true values of the 35 objective function coefficients are assumed to be those used in the work by *Hobbs* [1984]. The assumed errors are lognormal, with standard deviations equal to 40% of their respective true parameters. Errors in water use coefficients are assumed to be perfectly correlated. Coefficient errors for water sources of the same type (e.g., the six groundwater aquifers) are assumed to be highly correlated; however, the errors of surface and groundwater supply costs are assumed to be uncorrelated. We also assume that errors in demand and supply parameters have zero correlation. This is reasonable, as estimates of water supply costs were obtained from Texas government sources, while demand benefits were based on Electric Power Research Institute projections.

First, we solve the model using the true coefficients, yielding an objective function value $z(\mathbf{X}^*(\mathbf{C}_T)|\mathbf{C}_T)$ of \$9.7 $\times 10^6$ for the true optimum. We then use Monte Carlo simulation to generate fifty models with coefficients having the above described error distributions. Each model is solved, and the estimated objective function recorded. We then evaluate each solution using the assumed true coefficients, and note its true objective function value. Finally, we compare the estimated and true values to see if there is any optimistic bias.

The mean estimated objective function $z(\mathbf{X}^*(\hat{\mathbf{C}})|\hat{\mathbf{C}})$ of the 50 problems is \$11.9 $\times 10^6$, while the mean of the true values for the same solutions ($z(\mathbf{X}^*(\hat{\mathbf{C}})|\mathbf{C}_T)$) is \$8.4 $\times 10^6$. Figure 2 summarizes these results with a histogram of the differences between these values. The mean difference between the estimated and actual values, \$3.5 $\times 10^6$, had a standard error of \$1.3 $\times 10^6$, which makes it significantly larger than zero. Thus the optimistic bias is significant.

Equation (7) shows that an optimistic bias occurs in a maximization problem only if the coefficients and the optimal values of their variables are positively correlated. That was the case here. For each variable, except those whose values are always zero, the correlation is positive. The mean r is 0.49.

The implications of an optimistic bias in this type of analysis can be important. Imagine that the potential benefits of several different emerging technologies are being estimated by an optimization method of the above type. Because of objective function errors, it is possible that a subset of those will appear to have significant net benefits, which are likely to be biased upwards due to the bias identified here. These overly optimistic results may encourage a research funding agency to devote most of its research monies to these technologies, to the detriment of the others, and would also cause the agency to unwittingly exaggerate the potential benefits of its research. The real resource losses that can occur include the investment of too much research money into too small a subset of technologies whose promise is actually smaller than the optimization model says. A possible example of of this phenomenon is the devotion of most of the U.S. Department of Energy's research budget in the 1970s to nuclear breeder and fusion reactors, at the expense of fossil fuel, renewable, and conservation technologies.

In the present case, the results of our model were an input to the Electric Power Research Institute research planning process. If objective function values are, on average, optimistic, this could translate, in turn, into an exaggerated estimate of the worth of new cooling technologies, relative to existing, well-known technologies. An optimistic bias might encourage investment in technology development where none is actually justified.

CONCLUSION

We have argued by proof and example that there is reason to suspect that optimization is optimistically biased. Specifically, when objective function coefficients are subject to random estimation error, even if the error is unbiased, and errors in constraint sets are relatively unimportant, then the estimated value of the objective function is generally better, in an expected value sense, than its true value. Thus when a chosen alternative is implemented, its results will, on average, disappoint.

The expected size of this bias may in many cases be dwarfed by the overall uncertainty in the objective function, in which case it may be of no practical import. But in our Monte Carlo simulations, we found that the bias is of the same order of magnitude as the standard deviation of the objective. In the water for power model, as an example, the standard deviation of the objective was $\$8.7 \times 10^6$, compared to a mean bias of $\$3.5 \times 10^6$. The latter figure is one third the total net benefits of the assumed true optimum.

We therefore recommend that objective function values yielded by optimization models be taken with a grain (if not a shaker full!) of salt. One should be particularly careful when comparing the estimated performance of a solution with the performance of a water system already in place; the former may be overstated compared to the latter. This is especially true when coefficient estimates are untrustworthy, due to lack of information, inadequate sample sizes, or analytical difficulties. In a nutshell, "beware the low bidder."

Ideally, decision theoretic models that explicitly include posterior distributions of the true parameter values would be our salvation. Yet use of such models for the larger mathematical programs that arise in water planning is impractical. Analysts should therefore (1) conduct sensitivity analyses to see if model solutions are still attractive under a wide range of parameter values and (2) subject these solutions to simulation and modification, using independent sources of information to verify the cost and benefit estimates used in the model. Increased emphasis should be placed on verifying objective function coefficients using independent data sources. Of course, these recommendations are not new and, to a large extent, already constitute accepted practice; the possible presence of bias merely reinforces their importance. But if final design evaluations rely on the same data sources as the optimization model, then biases that arise in optimization might persist undiminished through the ensuing analyses.

APPENDIX

When errors in constraint sets are important, conclusions about the optimism of optimization become more ambiguous. The purpose of this appendix is to explore the effect of errors in constraint sets and to show that they do not necessarily yield either optimism or pessimism.

Consider first the LP below in which the $C_j > 0$, and a_j and b are scalars:

$$\text{Maximize } \Sigma_j C_j x_j \tag{A1}$$

subject to

$$\Sigma_j a_j x_j = b \qquad x_j \geq 0, \ \forall j \tag{A2}$$

In the optimum solution, only one of the x_j will be greater than zero. If only the right-hand side b is subject to error and not the objective function coefficients, no bias results. This (!) assumes that when the chosen solution is implemented, only the true b of the scarce resource is available and it is possible to adjust the chosen x_j so that amount and only that amount is used. Alternative assumptions about what "recourse" [*Dantzig and Madansky*, 1961] is possible could lead to different conclusions.

On the other hand, if the left-hand side coefficients a_j are the only ones subject to error, it can be shown that an (!) optimistic bias can result. The x_j that will be chosen will be the one which maximizes $bC_j/\hat{a}_j$. If it is assumed that the chosen x_j can later be increased or decreased to accommodate the true a_j, once known, then an optimistic bias results even if the true best alternative (the one with the highest bC_j/a_{Tj}) is always chosen. This is proven as follows. Let $x_j^*(\mathbf{a}_T)$ be the optimum under the true coefficients and let $\hat{a}_j^*$ be its estimated coefficient. If $x_j^*(\mathbf{a}_T)$ is optimal under all possible $\hat{\mathbf{a}}$, then the expection of the estimated objective function $E(z(\mathbf{X}^*(\hat{\mathbf{a}})|\hat{\mathbf{a}}))$ is $E(bC_j/\hat{a}_j^*) = bC_j E(1/\hat{a}_j^*)$. Since $1/\hat{a}_j^*$ is a strictly convex function, we then have

$$bC_j E(1/\hat{a}_j^*) > bC_j/\hat{a}_{Tj} = z(\mathbf{X}^*(\hat{\mathbf{a}})|\mathbf{a}_T) \tag{A3}$$

The inequality results if errors have a nonzero variance and are unbiased, and the second equality results from assuming that $x_j^*(\mathbf{a}_T)$ can be adjusted costlessly when the true $a_j^* = a_{Tj}^*$ is revealed. Thus there is an optimistic bias.

On the other hand, if the C_j in (A1) are negative, the result can be pessimistic bias. The following problem can also yield a pessimistic bias if only the right-hand sides are subject to unbiased error and the C_j are positive:

$$\text{Max} \sum_{j=1}^{2} C_j x_j \tag{A4}$$

subject to

$$0 \leq x_j \leq b_{jk} \qquad j = 1, 2 \qquad k = 1, \ldots, K_j > 2 \tag{A5}$$

In this problem, each x_j is subject to K_j upper bounds. If the recourse upon learning the true coefficients is to adjust the chosen x_j at no cost, then a pessimistic bias is likely. This is because errors in the $\hat{b}_{jk}$ are likely to result in pessimistic estimates of how much of each x_j is feasible. In most cases, once the true b_{jk} are known, it will be possible to implement more of the chosen x_j than initially believed possible.

Other formulations of problem (A4) and (A5) could lead to optimism instead. For example, $K_j = 1$, $\forall\ i$ results in an optimistic bias. Our point is that errors in constraint sets do not necessarily cause either optimism or pessimism.

NOTATION

- A set of realizations of $\hat{\mathbf{C}}$ such that $\mathbf{X}^*(\mathbf{C}_T)$ is still optimal under $\hat{\mathbf{C}}$.
- $\mathbf{a}_T$ vector of constraint coefficients, with elements a_j.
- $\hat{a}_j$ estimate of constraint coefficient for decision variable x_j.
- a_{Tj} true value of constraint coefficient for decision variable x_j.
- a_j^* constraint coefficient for optimal x_j.
- B set of realizations of $\hat{\mathbf{C}}$ such that $\mathbf{X}^*(\mathbf{C}_T)$ is not optimal under $\hat{\mathbf{C}}$.
- b right-hand side of constraint.
- $\hat{b}_{jk}$ estimated value of upper bound number k for decision variable x_j.
- $\hat{\mathbf{C}}$ vector of objective function coefficient estimates, with elements $\hat{C}_i$.
- $\mathbf{C}_T$ vector of true values of objective function coefficients, with elements C_{Ti}.
- $\mathbf{e}$ vector of unbiased estimation errors for objective function coefficients.
- e_{jk} estimation error for attribute j for alternative k.
- $E(\ )$ expectation operator.
- $f(\ )$ probability density.
- F feasible region, in decision space.
- K_j number of upper bounds upon decision variable x_j.
- $\mathbf{X}$ vector of decision variables.
- $\mathbf{X}^*(\mathbf{C})$ optimal solution under objective function coefficients $\mathbf{C}$.
- $\mathbf{X}^+(\mathbf{C})$ optimal solution to stochastic optimization problem under coefficients $\mathbf{C}$.
- $u_j(\ )$ single attribute utility function for attribute j.
- $U(\mathbf{X}_k)$ overall utility of alternative $\mathbf{X}_k$.
- w_j weight of attribute j.
- x_{jk} value of attribute j for alternative k.
- $\hat{x}_{jk}$ estimated value of attribute j for alternative k.
- x_{Tjk} true value of attribute j for alternative k.
- z objective function.
- $z(\mathbf{X}\mid\mathbf{C})$ value of objective function for solution $\mathbf{X}$ under objective function coefficients $\mathbf{C}$.
- $z(\mathbf{X}^*(\mathbf{C}_a)\mid\mathbf{C}_b)$ value of objective function z, calculated using coefficients C_b, of the optimal solution $\mathbf{X}^*$ obtained using $\mathbf{C}_a$.
- $z_i(\mathbf{X})$ term i of the objective function.

Acknowledgments. Helpful suggestions by V. Chankong, H. Chizeck, H. Ellis, M. Fiering, D. P. Loucks, H. Ellis, D. Maidment, J. Stedinger, H. Thomas, and an anonymous referee led to significant improvements in the paper. The authors are grateful to R. Krzysztofowicz, who pointed out that a correct decision theoretic formulation erases the bias. However, the responsibility for any remaining errors or opinions is solely ours. Funds for this research were provided by the National Science Foundation under grant ECE-85-52524.

REFERENCES

Bard, J. F., and S. Chatterjee, Objective function bounds for the inexact LP problem with generalized cost coefficients, *Comp. Oper. Res.*, *12*(5), 483–491, 1985.

Bell, D. E., Regret in decision making under uncertainty, *Oper. Res.*, *20*, 961–981, 1982.

Bell, D. E., Disappointment in decision making under uncertainty, *Oper. Res.*, *33*(1), 1–27, 1985.

Dantzig, G. B., and A. Madansky, On the solution of two-stage linear programs under uncertainty, in *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, vol. I, pp. 165–176, University of California Press, Berkeley, 1961.

Esmaeil-Beik, S., and Y-S. Yu, Optimal operation of multipurpose pool of Elk City lake, *J. Water Resour. Plann. Manage.*, *110*(1), 1–14, 1984.

Friedman, R., C. Ansell, S. Diamond, and Y. Y. Haimes, The use of models for water resources management, planning, and policy, *Water Resour. Res.*, *20*(7), 793–802, 1984.

Gershon, M., L. Duckstein, and R. McAniff, Multiobjective river basin planning with qualitative criteria, *Water. Resour. Res.*, *18*(2), 193–202, 1982.

Hobbs, B. F., Water supply for power in Texas-Gulf region, *J. Water Resour. Plann. Manage.*, *110*(4), 373–391, 1984.

Hobbs, B. F., and P. M. Meier, An analysis of water resources constraints on power plant siting in the mid-Atlantic states, *Water Resour. Bull.*, *15*(6), 1666–1676, 1979.

Keeney, R. L., and E. F. Wood, An illustrative example of the use of multiattribute utility theory for water resource planning, *Water Resour. Res.*, *13*(4), 705–712, 1977.

Itami, H., Expected objective value of a stochastic linear program and the degree of uncertainty of parameters, *Manage Sci.*, *21*(3), 291–301, 1974.

Krzysztofowicz, R., Why should a forecaster and a decision maker use Bayes theorem, *Water Resour. Res.*, *19*(2), 327–336, 1983.

Liebman, J. C., Some simple-minded observations on the role of optimization in public systems decision making, *Interfaces*, *6*(4), 103–107, 1976.

Leighton, J. P., and C. A. Shoemaker, An integer programing analysis of the large wastewater treatment and collection systems, *Water Resour. Res.*, *20*(6), 671–681, 1984.

Loucks, D. P., J. Stedinger, and D. Haith, *Water Resource Systems Planning and Analysis*, Prentice-Hall, Englewood Cliffs, N.J., 1981.

Loucks, D. P., J. R. Stedinger, and U. Shamir, Research in water resources and environmental systems modeling: Some historical perspectives, current issues, and future directions, *Natl. Resour. Forum*, *8*(3), 210–230, 1984.

Mangasarian, O. L., Nonlinear programming problems with stochastic objective functions, *Manage. Sci.*, *10*, 353–359, 1964.

Raiffa, H., and R. Schlaifer, *Applied Statistical Decision Theory*, Harvard University, Boston, Mass., 1961.

Read, E. G., and J. F. Boshier, Biases in stochastic reservoir scheduling models, paper presented at Bellman Workshop on Dynamic Programming in Water Resources, Nat. Sci. Found., Atlanta, Ga., June 1986.

Rogers, P. P., On the choice of the "appropriate model" for water resources planning and management, *Water Resour. Res.*, *14*(6), 1003–1010, 1978.

Rogers, P. P., Role of systems analysis as a tool in water policy planning and management, Water resources planning experiences in a national and regional context, *Rep. TCD/SEM.80/1*, United Nations Dep. of Tech. Coop. for Devel., New York, 1980.

Rogers, P. P., and M. B. Fiering, Use of systems analysis in water management, *Water Resour. Res.*, *22*(9), 146S–158S, 1987.

Thomas, H. A., Jr., Capacity expansion of public works, report, 14 pp., Harvard Univ., Cambridge, Mass., January 1967.

Wets, R. J.-B., The distribution problem and its relation to other problems in stochastic programming, in *Stochastic Programming*, edited by M. A. H. Dempster, pp. 245–262, Academic, San Diego, Calif., 1980.

Wyatt, T., P. E. Robinson, and T. C. Muir, Systems simulation and optimization—Useless, useful, or essential? in *Proceedings, Sixth World Congress of the International Water Resources Association, Ottawa*, International Water Resources Association, Urbana, Ill., 1988.

A. Hepenstal, 6250 Center Hill Road, Procter & Gamble Corporation, Cincinnati, OH 45224.

B. F. Hobbs, 610 Crawford Hall, Case Western Reserve University, Cleveland, OH 44106.

(Received November 11, 1986;
revised August 2, 1988;
accepted September 19, 1988.)